

DANIEL LIAN

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🐙 github.com/GlutenFreeGrapes

Highly motivated graduate student in Computer Science with passion for software engineering, artificial intelligence, machine learning, cybersecurity, and data science. Outside of computer science, I enjoy running and volleyball.

Education

University of Michigan

Master of Science in Engineering in Computer Science & Engineering

Aug. 2026 – May 2027

Ann Arbor, Michigan

University of Michigan

Bachelor of Science in Engineering in Computer Science - Summa Cum Laude (3.76/4.00)

Aug. 2023 – May 2026

Ann Arbor, Michigan

Relevant Coursework

- | | | | |
|----------------------------------|---------------------|-----------------------------------|-----------------------------|
| • Data Structures and Algorithms | • Data Science | • Foundations of Computer Science | • Operating Systems |
| • Software Design | • Conversational AI | • Discrete Math | • Computer Organization |
| • Problem Solving | • Ethics and AI | • Computer Vision | • Computational Linguistics |
| | • Computer Security | | |
| | • Cryptography | | |

Experience

Fairfax Collegiate

June 2023 – August 2025

Head Instructor

McLean, VA

- Led 15 STEM classes over 15 weeks, teaching over 100 students from 3rd grade through 10th grade robotics, Scratch programming, advanced math concepts (including middle school math and algebra), and game design.
- Honed communication skills by mentoring junior staff, created friendlier team-based learning environment to adapt to students' diverse learning styles.

Projects

Compas Course Recommender | *TypeScript, HTML, PostgreSQL, Python, Selenium*

January 2026 - May 2026

- Designed and developed system using Gemini backend LLM to provide University of Michigan course recommendations and reasoning based on user's degree program, course history, and preferences with iterative feedback mechanism.
- Used PostgreSQL backend to store course data from over 14,000 courses with metrics for interest and workload.

Cities Chain Discord Bot | *Python, Pandas, MariaDB, SQL*

July 2023 - Present

- Designed and implemented a scalable Discord game supporting thousands of players across 700+ Discord servers using efficient SQL backend data management in MariaDB for seamless gameplay.
- Applied Agile software maintenance principles and used player feedback to ensure robust gameplay experience.

Subway Challenge | *Python, Pandas*

December 2022 - May 2024

- Collected and analyzed GTFS data for metropolitan rail systems across 6 cities (Kochi, Los Angeles, Atlanta, San Francisco, Washington, New York), created graph structure using schedule data to model regular routes.
- Implemented heuristics-based breadth-first search algorithm to find fastest complete traversals of all metro systems.

Election Prediction Algorithm | *Python, Jupyter Notebook, TensorFlow, Keras, Pandas, NumPy*

August 2022 - May 2023

- Implemented Python Tensorflow model in Jupyter Notebook to predict 85% of 2022 United States congressional winners using weighted average of 538 polls, demonstrating robust predictive analytics capabilities.
- Presented research and findings as part of senior research project to audience of 40 at tjSTAR technology research symposium.

Technical Skills

Languages: Python (Pytorch, TensorFlow, Selenium, Flask, Pandas, NumPy, scikit-learn), C/C++, Java, SQL, HTML/CSS, JavaScript, LaTeX

Developer Tools/Technologies/Frameworks: VS Code, Eclipse, Linux, Git, GitHub, MariaDB, Jupyter Notebook

Certificates: Amazon Web Services Certified Solutions Architect Associate

Leadership / Extracurricular

MHacks 2025

September 2025

Participant

University of Michigan

- Designed and developed Spanish-language learning chatbot using Typescript frontend and Python backend
- Implemented tokenization/dictionary functionality for easy vocabulary learning, used 2-model system consisting of summarizer model and chatbot model to reduce token usage and store facts from previous conversations

Extracurriculars: Asian Interscity, WolvSec, Michigan Quizbowl, 3D Printing Club